



Structural brain network deviations predict recovery after traumatic brain injury

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ABSTRACT

Objective: Traumatic brain injury results in diffuse axonal injury and the ensuing maladaptive alterations in network function are associated with incomplete recovery and persistent disability. Despite the importance of axonal injury as an endophenotype in TBI, there is no biomarker that can measure the aggregate and region-specific burden of axonal injury. Normative modeling is an emerging quantitative case-control technique that can capture region-specific and aggregate deviations in brain networks at the individual patient level. Our objective was to apply normative modeling in TBI to study deviations in brain networks after primarily complicated mild TBI and study its relationship with other validated measures of injury severity, burden of post-TBI symptoms, and functional impairment.

Method: We analyzed 70 T1-weighted and diffusion-weighted MRIs longitudinally collected from 35 individuals with primarily complicated mild TBI during the subacute and chronic post-injury periods. Each individual underwent longitudinal blood sampling to characterize blood protein biomarkers of axonal and glial injury and assessment of post-injury recovery in the subacute and chronic periods. By comparing the MRI data of individual TBI participants with 35 uninjured controls, we estimated the longitudinal change in structural brain network deviations. We compared network deviation with independent measures of acute intracranial injury estimated from head CT and blood protein biomarkers. Using elastic net regression models, we identified brain regions in which deviations present in the subacute period predict chronic post-TBI symptoms and functional status.

Results: Post-injury structural network deviation was significantly higher than controls in both subacute and chronic periods, associated with an acute CT lesion and subacute blood levels of glial fibrillary acid protein ($r = 0.5$, $p = 0.008$) and neurofilament light ($r = 0.41$, $p = 0.02$). Longitudinal change in network deviation associated with change in functional outcome status ($r = -0.51$, $p = 0.003$) and post-concussive symptoms (BSI: $r = 0.46$, $p = 0.03$; RPQ: $r = 0.46$, $p = 0.02$). The brain regions where the node deviation index measured in the subacute period predicted chronic TBI symptoms and functional status corresponded to areas known to be susceptible to neurotrauma.

Conclusion: Normative modeling can capture structural network deviations, which may be useful in estimating the aggregate and region-specific burden of network changes induced by TAI. If validated in larger studies, structural network deviation scores could be useful for enrichment of clinical trials of targeted TAI-directed therapies.

Abbreviations: BSI, Brief Symptom Inventory-18; RPQ, Rivermead Post-Concussion Symptom Questionnaire; CT, Computed Tomography; MRI, Magnetic Resonance Imaging; TAI, Traumatic Axonal Injury; TBI, Traumatic Brain Injury.

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1. Introduction

Traumatic brain injury (TBI) is a global public health issue, and even its mildest forms—mild TBI (mTBI) or concussion—carries a significant risk of chronic neurological complications such as dementia (Barnes et al., 2018) and epilepsy (Pugh et al., 2015; Pugh et al., 2021). TBI results in injury to neurons, glia, and cerebral blood vessels with substantial variability in severity and anatomic location between individuals. Injury can also be conceptualized as resulting in focal and diffuse brain insults (Sharp et al., 2014). This mechanistic and spatial heterogeneity likely plays a large role in the substantial inter-individual variability in outcomes. Although acute CT and MRI findings such as the

presence of traumatic subarachnoid hemorrhage and contusions influence outcome after TBI (Yuh et al., 2013), efforts at developing targeted therapies for preventing or mitigating the most severe complications of TBI, such as dementia and epilepsy (Diaz-Arrastia et al., 2014) requires more sophisticated methods of risk stratification.

The most common and well-described pathologic feature of TBI is traumatic axonal injury (TAI) (Smith et al., 2003; Johnson et al., 2013), TAI is responsible for much of the long-term disability after neuro-trauma (Johnson et al., 2013) and may incite the process of progressive neurodegeneration (Graham et al., 2020). Disability resulting from TAI may be a consequence of injury to critical axonal tracts, such as brainstem reticular pathways (Edlow et al., 2013), but more commonly

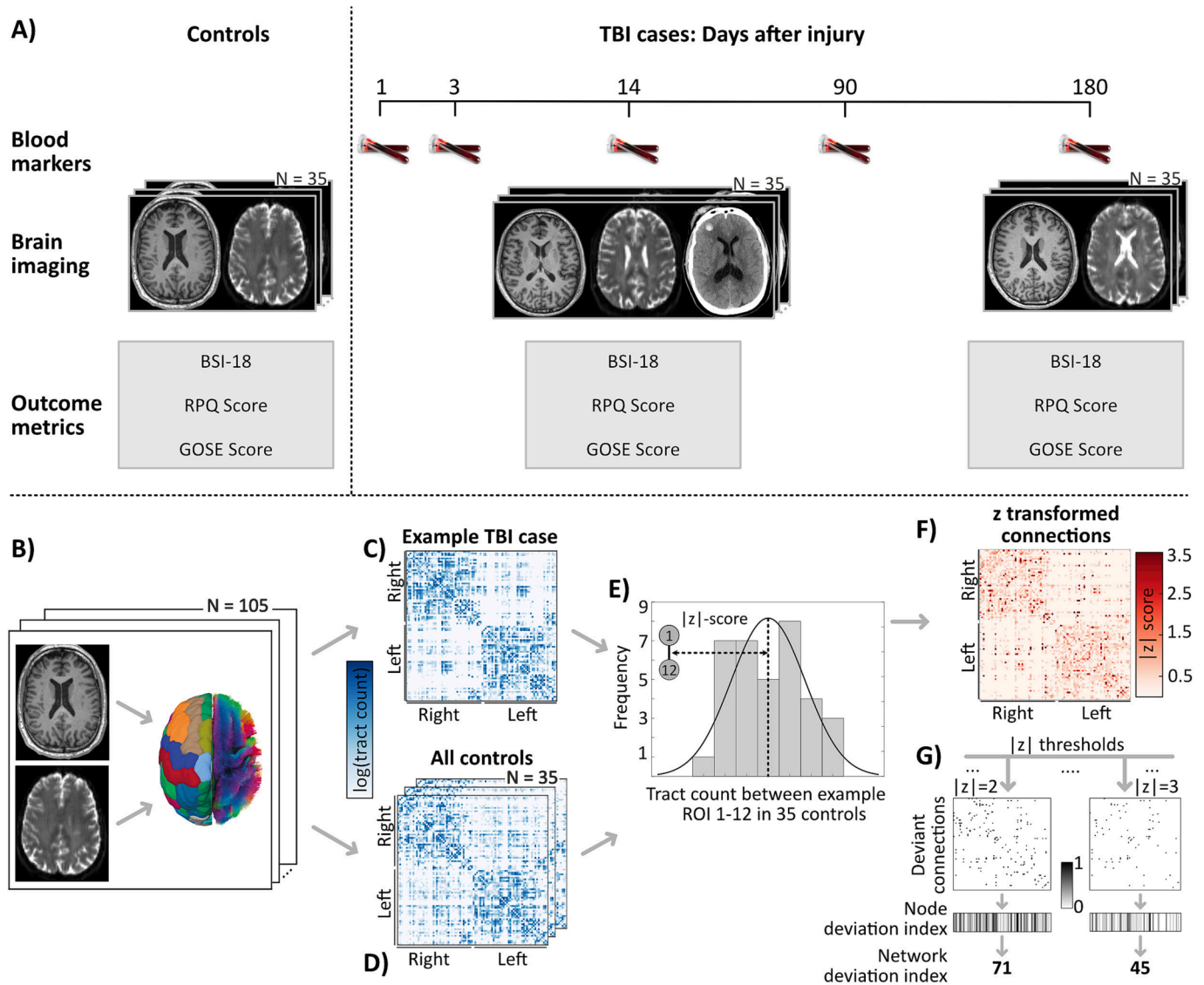


Fig. 1. Overall approach. (A) Brain imaging, blood draws, and outcome assessment data for post-TBI were collected for 35 controls and 35 TBI participants. The subacute period was approximately 14 days and the chronic period was approximately 180 days (6 months) after injury. Brain imaging included the T1-weighted and diffusion-weighted MRI for all controls and individuals with TBI during the subacute and chronic periods. CT images were acquired for all TBI cases upon presentation to the hospital. Blood plasma concentrations of GFAP and NFL (blood biomarkers of TBI) were estimated at different time points after injury (only subacute and chronic periods were analyzed as these aligned with imaging acquisition). (B) For each brain imaging scan across all individuals, we constructed a structural brain network consisting of gray matter regions/nodes and the white matter streamlines connecting each region/node. (C, D) We generated a connectivity matrix for each scan. The x- and y-axes of the matrix are the nodes identified by the brain parcellation scheme, and the elements of the matrix are the connections representing the number of white matter streamlines connecting each pair of nodes. (E, F) We computed the z-score for every connection by normalizing each connection with the mean and standard deviation of the corresponding connections in uninjured controls. The normalization transformed the connectivity matrix into a z-score matrix, where a high value indicated a more deviant connection. (G) We assessed deviant connections at multiple z-score thresholds. We binarized the z-transformed connections at each threshold, with one indicating a deviant connection. Node deviation index is the ratio of deviant connections to a node/region over the total number of connections to that node. Network deviation index is the total number of deviant nodes.

is associated with the aggregate burden of disconnection in widely distributed neural networks (Johnson et al., 2013; Wang et al., 2011). Indeed, some white matter tracts are more vulnerable than others due to their orientation and location, such as the midline structures (corpus callosum, fornix, and cingulum) or the gray-white junction between tissue compartments (Wang et al., 2011; Wang et al., 2008; Cloots et al., 2013; Cloots et al., 2008). While TAI was initially described in the setting of severe TBI and prolonged coma, it is now widely recognized that TAI is a predominant mechanism of disability in mild and moderate TBI as well (Johnson et al., 2013). Unfortunately, TAI is frequently undetectable using clinically available methods and is therefore likely to be largely underestimated in clinical practice.

TAI can be characterized *in vivo* by diffusion MRI, with most studies reporting a disruption of white matter microstructural integrity in TBI (Wallace et al., 2018). Numerous methods exist for the analysis of diffusion MRI data; however, current analytic methods generally measure group effects where group differences (i.e., comparison of group averages) are assessed with a separate statistical test at each of the hundreds of thousands of voxels or vertices in the brain. With this experimental design, the groups are assumed to be homogenous; however, the assumption of within-group homogeneity is particularly problematic in TBI where brain pathology may be multifocal and spatially heterogeneous between individuals and hence not well represented by a regional group average. Spatial normative modeling is an alternative, data-driven technique for capturing biologically relevant subject-level differences in disorders with spatially heterogeneous brain pathology. Spatial normative modeling involves estimation of a reference distribution for a given regional neuroimaging measure, which is derived from a control population. The extent to which a patient deviates from the control population is then estimated for each region and these values are used to generate a brain map of regional deviation scores. The deviation scores can be summed to give an aggregate subject-level measure of overall deviation. Thus, spatial normative modeling is a promising case-control approach that quantifies abnormalities in individual cases as the deviations from a matched cohort of healthy individuals (Verdi et al., 2021; Marquand et al., 2019). Recent application of normative modeling characterized both the whole brain as well as region-specific structural network abnormalities in Alzheimer's disease (Verdi et al., 2021) and epilepsy (Sinha et al., 2021a,b). Here, we introduce the concept of structural network deviation, a measure of the overall burden of structural network abnormalities and node deviation index, a readout of the burden of abnormalities at individual network nodes. This approach is a promising tool for characterizing the overall burden and spatial pattern of TAI. An aggregate measure of TAI would be useful as a biomarker in future therapeutic trials of agents targeting TAI either for patient enrichment or as a pharmacodynamic biomarker of treatment effect. Tracking changes in the aggregate burden of TAI may be useful as a prognostic biomarker by identifying individuals who may not fully recover after TBI, as it can be tracked over time to monitor disease progression.

The main objective of this study was to describe a novel method for assessment of axonal integrity—network deviation index—derived from normative modeling of structural connectivity data in a cohort of patients with primarily complicated mild TBI imaged at two time points in the subacute and chronic post-injury period. We hypothesized that the network deviation index would correlate with validated measures of injury severity, burden of post-TBI symptoms, and functional impairment. To test this hypothesis, we quantified the network deviation index inferred from diffusion MRI data acquired during the subacute and chronic post-injury period by applying a spatial normative modeling approach. We compared this proxy measure of traumatic axonal injury to two complementary modalities used clinically to assess injury severity: a) abnormalities on CT images and b) expression of blood protein biomarkers of axonal and glial injury. We found network deviation index measured in the subacute period predicts functional outcome and burden of post-injury symptoms in the chronic period.

2. Materials and methods

2.1. Participant characteristics

We recruited adults (age 18 and older) hospitalized for non-penetrating TBI at a single Level 1 Trauma Center. Individuals with a history of pre-existing serious neurological or psychiatric disorder, comorbid disabling condition limiting outcome assessment, current pregnancy, those under legal custody, or under the age of 18 were excluded. Thirty-five demographically matched healthy control subjects with no history of TBI, pre-existing disabling neurological or psychiatric disorder, or current pregnancy were recruited from the general population via local advertisement. Individuals with TBI underwent brain MRI at two weeks and six months post-injury period as well as blood draws at multiple time points, including two weeks and six months post-injury. Healthy controls underwent a single brain MRI. Demographic information, medical history, admission injury characteristics, and other clinical information were collected from the medical records. These details and timelines are illustrated in Fig. 1a. This study was approved by the Institutional Review Board at the University of Pennsylvania, and written consent was obtained by the individual or a legally authorized representative when appropriate.

2.2. Acute head CT interpretation

Upon presentation to the Emergency Department, each TBI participant underwent a CT scan to assess for the presence of an acute post-traumatic intracranial lesion. Each participant's CT scan and the accompanying clinical neuroradiologist's report were independently evaluated by a board-certified neurologist (JG) for the presence or absence of acute intracranial lesions. An acute intracranial lesion was defined as any trauma-related finding, including epidural hematomas, subdural hematomas, indeterminate extra-axial lesions, contusions, intraventricular hemorrhage, subarachnoid hemorrhage, and petechial haemorrhage (Yuh et al., 2021). Presence of any of the above intracranial lesions was considered CT-positive; otherwise, the individual was categorized as CT-negative. Consistent with prior studies (Yuh et al., 2021), isolated skull fracture was not considered an acute intracranial lesion.

2.3. Measurement of neurodegeneration-associated proteins in blood serum

Neurofilament light (NfL) is a protein component of the neuronal cytoskeleton and a putative biomarker of axonal injury that is released into the CSF and blood in several neurodegenerative diseases, including TBI (Bacioglu et al., 2016). Glial fibrillary acidic protein (GFAP) is a protein expressed almost exclusively in astrocytes and is released into the CSF and blood following the disintegration of the astrocyte cytoskeleton (Diaz-Arrastia et al., 2014). Blood concentrations of both proteins increase acutely after brain injury and are associated with neuroimaging abnormalities (Shahim et al., 2020a,b) and outcomes after TBI (Lei et al., 2015; Vos et al., 2010). To establish the relationship between structural network abnormalities and blood markers of TAI, we collected blood samples via venipuncture within 24 h, 72 h, two weeks, three months, and six months of injury. NfL and GFAP were measured on the Quanterix HD-X Analyzer™ (Quanterix, Billerica, MA) via Single Molecule Assay (Simoa®), using the 'Neurology 4-plex B' immunoassay kits.

2.4. Post-injury recovery assessment

Post-injury recovery was assessed at the same time as the MRI using three measures: self-reported symptomatology was assessed with the Rivermead Post-Concussion Symptom Questionnaire (RPQ) (King et al., 1995) and the Brief Symptom Inventory-18 (BSI-18), while the post-

injury functional outcome was assessed using the Glasgow Outcome Scale-Extended (GOSE) (Jennett et al., 1981). Outcome assessment was conducted by trained research personnel who were blind to imaging and biomarker findings. To understand the relationship between the temporal profile of neuroimaging abnormalities and post-TBI symptom burden, we derived a measure that includes information at both time points as well as their change over time. This is defined as proportional change. For measures of symptom severity, the proportional change is defined as the following:

$$\text{change in outcome score} = \frac{\text{score (chronic)} - \text{score (subacute)}}{\text{score (chronic)} + \text{score (subacute)}}$$

2.5. Neuroimaging data acquisition and preprocessing

Brain MRIs were performed on a 3 T scanner (Siemens Prisma) using a product 32-channel head coil. Structural imaging included a sagittal T1-weighted MPRAGE (TR = 2.3 s, TE = 2.94 ms, TI = 900 ms, FA = 9°, resolution = 1 × 1 × 1 mm). Whole brain diffusion MRI was performed with an echo planar sequence with FA 90° and resolution = 2.4 × 2.4 × 2.4 mm (b-value = 1000 s/mm², 64 diffusion directions, TR 2.9 s, TE 94 ms). Separately, 14 images with b-value = 0 s/mm² with reverse phase encoding were acquired. Diffusion MRI data were corrected for eddy current and movement artifacts using the FSL topup and eddy tool (Jenkinson et al., 2012). The diffusion data for each subject was registered and reconstructed to the standard ICBM-152 space using q-space diffeomorphic reconstruction implemented in DSI studio (<https://dsi-studio.labsolver.org>). We visually inspected all data to assess for inaccuracies in image registration and tract reconstruction in the presence of focal lesions. We did not identify any cases with errors in registration or tract reconstruction. Focal post-traumatic lesions did not affect registration quality, with $r^2 = 0.83 \pm 0.01$ between native and standard space template MRI.

2.6. Measurement of structural network deviations

The process for delineating structural network deviation is outlined in Fig. 1. For each participant, we constructed a structural brain network consisting of gray matter regions (nodes) and white matter connections (edges) between each region. We defined 90 contiguous cortical and subcortical nodes from the AAL atlas. To identify the connectivity between regions, we performed whole-brain fiber tracking using deterministic tractography. The choice of fiber tracking algorithm (deterministic or probabilistic) depends on whether sensitivity or specificity is more critical to the analysis (Maier-Hein et al., 2017). A recent study demonstrated that false positive connections can significantly impact network measures obtained from connectome analysis methods, highlighting the importance of minimizing these false connections in our analysis (Zalesky et al., 2016). Therefore, as in our previous studies, we opted for a deterministic fiber tracking algorithm informed by neuro-anatomical priors to reduce the occurrence of false positive connections and maintain the accuracy of our connectome analysis (Sinha et al., 2021a,b; Yeh et al., 2019; Yeh et al., 2018). We applied identical fiber tracking parameters on each scan across all subjects. Tractography generated approximately 1,000,000 tracts with tracking parameters configured as follows: Runge-Kutta method with step size 1 mm, whole-brain seeding, initial propagation direction set to all fiber orientations, minimum tract length 15 mm, maximum tract length 300 mm, and topology informed pruning applied with one iteration to remove false connections. The whole-brain structural network is constructed by delineating connections between all pairs of gray matter regions of interest (ROIs) in the parcellation scheme. Using this schema, a structural connection exists between two regions if they are linked by a streamline. We weighted the connectivity across all streamlines that connect each pair of regions by the average volume of the regions connected by those streamlines (Taylor et al., 2015). Repeating this process for each pair of

nodes resulted in weighted connectivity matrices of size 90 × 90 per participant, where the x- and y-axes of the matrix are the regions defined by the brain parcellation scheme, and the elements of the matrix are populated with the white matter streamlines connecting each pair of regions (Fig. 1b–d).

After structural network construction, we computed a z-score for each edge between regions in each TBI participant using the mean and standard deviation of the respective edge in the uninjured controls. Given that networks inferred from deterministic tractography are sparse, we z-scored only those edges in individuals with TBI for which an equivalent edge existed in at least 12 (~35 %) controls (de Reus and van den Heuvel, 2013; Jolly et al., 2021). An edge was considered deviant if the absolute value of the z-score exceeded a specific threshold. To identify the appropriate z-score threshold beyond which a node is considered deviant, we varied the z-score threshold from 2 to 3.5 in increments of 0.1. **Node deviation index** is the proportion of deviant edges divided by the total number of non-zero edges for a given node. **Network deviation index** is the total number of nodes in the network with at least one deviant connection. These steps are illustrated in Fig. 1b–f. To ensure robustness, we evaluated a range of z-score thresholds as well as with alternative (higher resolution) atlas of brain parcellation. In the main manuscript, we show the result at one example threshold at $z > 2.5$.

Similar to the proportional change for symptom severity, we calculated the proportional change of post-TBI network deviation index:

$$\text{change in network deviation} = \frac{\text{deviation (chronic)} - \text{deviation (subacute)}}{\text{deviation (chronic)} + \text{deviation (subacute)}}$$

2.7. Structural network model for recovery prediction

To identify structural network nodes important for recovery, we mapped the relationship between node deviation index among the 90 individual nodes in the subacute post-injury period with recovery measures in the chronic post-injury period, incorporating feature selection with elastic net regression (Zou and Hastie, 2005). Elastic net regression is a regularization technique that permits the identification of important predictors in the presence of numerous redundant or highly correlated predictors. Elastic net is a combination of lasso (L1 norm) and ridge (L2 norm) regression aiming to optimize the following objective function:

$$\min_{\beta_0, \beta} \left(\frac{1}{2N} \sum_{i=1}^N (y_i - \beta_0 - x_i^T \beta)^2 + \lambda P_\alpha(\beta) \right),$$

where,

$$P_\alpha(\beta) = \frac{(1 - \alpha)}{2} \|\beta\|_2^2 + \alpha \|\beta\|_1$$

In this objective function, N is the number of observations, y_i is the response at observation i , x_i is the input data or feature vector of length p at observation i , and parameters β_0, β are the scalar intercept and weight vector of length p which are estimated by the model training process. λ (lambda) and α (alpha) are the regularization parameters that scale the L1 ($\|\beta\|_1$) and L2 ($\|\beta\|_2^2$) norms to prevent overfitting of the regression model.

To capture the underlying driving force behind chronic post-traumatic symptoms associated with incomplete recovery, we applied principal component analysis to the scores of the BSI-18, RPQ, and GOSE obtained in the chronic post-injury period. As these three scales measure many of the same symptoms, the first principal component represents a large amount of the overall variance. The first principal component score was entered as the response variable (y_i) in the elastic net regression model. The node deviation index for each of the 90 network nodes measured in the subacute post-injury period was entered as a feature vector (x_i of length $p = 90$). During model training, parameter β

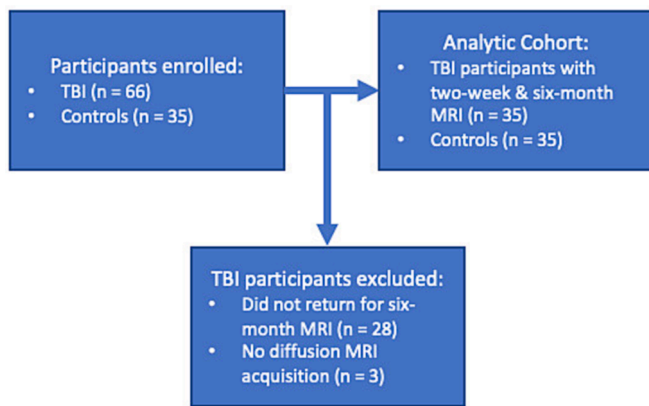


Fig. 2. Study Flow.

weighted each network node, which corresponded directly to the relative importance of the node deviation index of that node in predicting the recovery status (first principal component score) of a patient in the chronic post-injury period. The regularization parameters λ (lambda) and α (alpha) guided the feature selection process by identifying a minimum number of nodes from the 90 nodes (features) that increases the prediction performance of the regression model on held-out data. We incorporated a nested cross-validation scheme for model training, tuning, and testing. In the outer fold, we held out 10 % of the data to test the model prediction accuracy. In the inner fold, we incorporated a tenfold cross-validation scheme for model training and tuning. We trained the model on the training fold and tuned the model with different regularization parameters on the inner cross-validation folds. To select the appropriate regularization parameters, we set $\alpha = 0.5$, and performed a grid search varying λ over values ranging from 0.001 to 1 (Sinha et al., 2021a; Fagerholm et al., 2015). We selected the model with the lowest mean square error for the testing model prediction performance on unseen test data from the outer fold. The resultant model also included normalized coefficients indicating the relative importance of the individual nodes that predicted patient recovery. We performed a Monte Carlo simulation where this process was repeated 10,000 times and the median predicted value of the recovery score was extracted for comparison with the observed recovery score.

2.8. Statistical analysis

Data distributions were first visually assessed and then assessed via the Kolmogorov-Smirnov test for normality. Given the non-normal distribution of network deviation index, we applied the non-parametric Wilcoxon rank-sum test to assess for group level differences in network deviation index between TBI patients and controls. We utilized the ranksum function in MATLAB (MathWorks, Inc, Natick, MA), incorporating the exact method and a one-tailed p value. To assess change in network deviation index in TBI cases from subacute to chronic period, we applied paired statistics, testing significance using the signed ranksum test. We conducted these tests across a range of z-score thresholds from 2.0 to 3.5 (for a total of 45 statistical tests) and present results after false discovery rate correction using the Benjamini-Hochberg method with $p < 0.05$ considered statistically significant. Effect size between groups was computed with the Cohen's d score. We computed 95 % bootstrap confidence intervals (CIs) of effect size using a bias-corrected and accelerated percentile method from 10,000 bootstraps resamples with replacement. The correlation coefficients between blood biomarkers and network deviation index as well as between proportional change in network deviation index and measures of clinical outcomes were determined with the Spearman rank order. We did not impute missing values to avoid introducing any bias. Chi-squared or Fisher's exact test was used for comparing categorical data between

Table 1

		TBI (n = 35)	Controls (n = 35)	p-value
Age, median (IQR)		29.0 (23.0–43.0)	31.0 (23.0–34.0)	0.37
Male		27 (77 %)	22 (63 %)	0.30
Race	Black	13 (37 %)	6 (17 %)	0.26
	Other	4 (11 %)	5 (14 %)	
	White	18 (51 %)	21 (60 %)	
Hispanic		1 (3 %)	4 (11 %)	0.19
Educational category	< high school	3 (9 %)	0 (0 %)	0.19
	> high school high school or GED equivalent	22 (63 %) 8 (23 %)	23 (66 %) 4 (11 %)	
Smoker		10 (29 %)	1 (3 %)	1.00
AUDIT-C, median (IQR)		2.0 (0.0–4.0)	3.0 (2.0–5.0)	0.17
AUDIT-C category	heavy alcohol	11 (31 %)	7 (20 %)	0.33
	light/moderate alcohol	9 (26 %)	2 (6 %)	
	no alcohol	11 (31 %)	2 (6 %)	
Illicit drug use		7 (20 %)	1 (3 %)	1.00
Anxiety		1 (3 %)	1 (3 %)	0.52
Depression		5 (14 %)	1 (3 %)	0.64
Hypertension		2 (6 %)	0 (0 %)	1.00
Sleep disorder		0 (0 %)	1 (3 %)	0.30
Prior seizure disorder		0 (0 %)	1 (3 %)	0.28
Migraine		1 (3 %)	0 (0 %)	1.00
Prior TBI		4 (11 %)	0 (0 %)	0.55
Injury cause	fall	8 (23 %)	0 (0 %)	1.00
	non-intentional injury or violence/assault	8 (23 %)	0 (0 %)	
	road traffic incident	18 (51 %)	1 (3 %)	
Loss of consciousness		24 (69 %)	0 (0 %)	0.27
Post-traumatic amnesia		20 (57 %)	0 (0 %)	0.33
GCS, median (IQR)		15.0 (14.0–15.0)		
CT positive (i.e., complicated mild TBI)		22 (63 %)		
Contusion		12 (34 %)		
Epidural hemorrhage		3 (9 %)		
Subarachnoid hemorrhage		14 (40 %)		
Sudural hemorrhage		8 (23 %)		
Skull fracture		11 (31 %)		
Intraventricular hemorrhage		1 (3 %)		

Abbreviations: AUDIT-C, Alcohol Use Disorder Identification Test; GCS, Glasgow Coma Score; GOSE, Glasgow Outcome Scale Extended TBI, traumatic brain injury.

groups. We applied the elastic net model as implemented in the “lasso” function in MATLAB.

2.9. Data availability

De-identified clinical and imaging data will be made available upon reasonable request to the corresponding author.

3. Results

3.1. Baseline characteristics of study participants

Sixty-six participants with TBI and 35 controls were enrolled in the study. Twenty-eight TBI participants completed a two-week MRI but did not return for a six-month MRI. The analytic cohort included 35 TBI

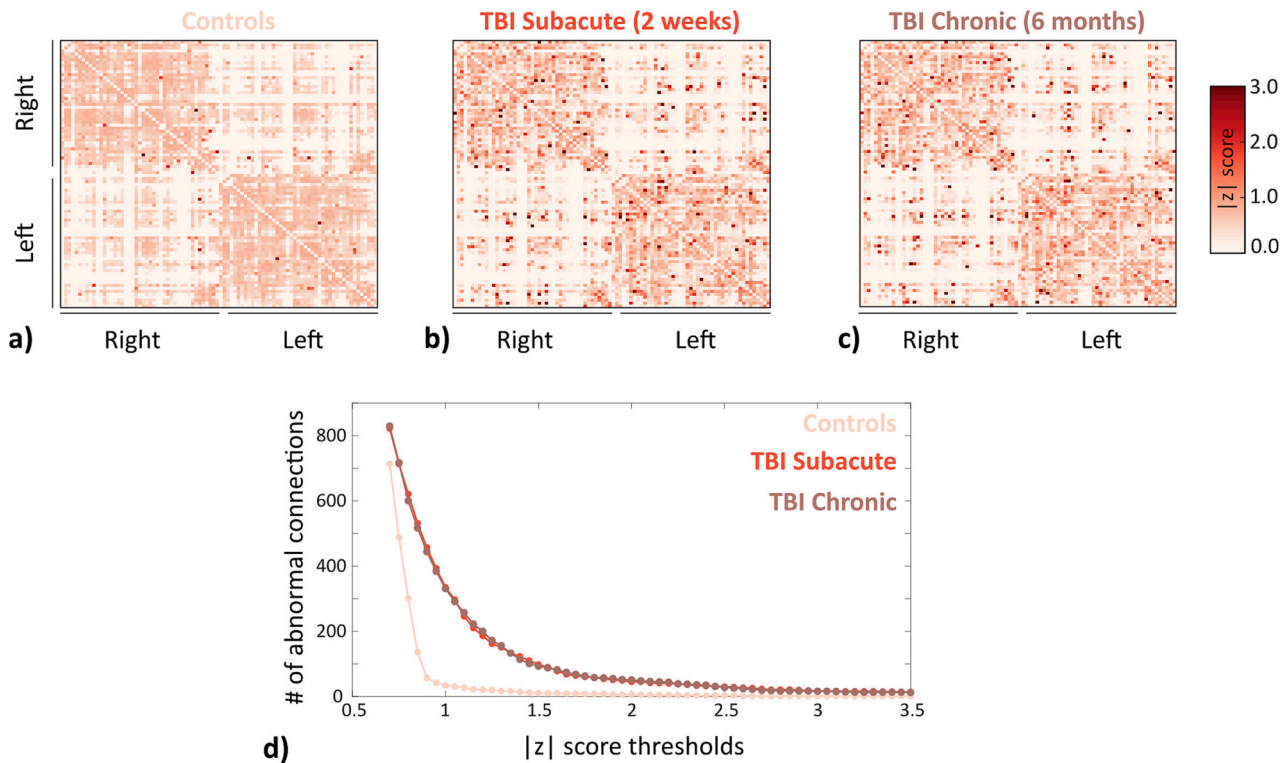


Fig. 3. Group level deviations in whole brain structural connectivity. The top row shows *average* adjacency matrices across controls (a), and TBI participants in the subacute (b) and chronic (c) post-injury period. The elements of each matrix are populated with the node-specific z-score derived from the mean and standard deviation of the respective node in uninjured controls (as illustrated in the overall approach Fig. 1f). Panel d) At the group level, participants with TBI have more abnormal connections than controls across a wide range of z-score thresholds. Panel d) shows the group level effect on abnormal connection count across a range of z-score thresholds. The y-axis denotes the total (sum) number of abnormal connections in the whole brain network for each cohort.

participants and 35 controls (Fig. 2). Study demographics and injury characteristics for the analytic cohort are shown in Table 1. All participants had Glasgow Coma Scale (GCS) between 13 and 15 (median GCS 15; interquartile range (IQR) 14–15). Twenty-two of 35 (63 %) had an acute intracranial lesion on CT and hence the cohort consisted primarily of those with complicated mild TBI. Study demographics and injury characteristics were similar for the analytic cohort and those who did not return for a six-month MRI (Supplementary Table).

3.2. Network deviation index among patients with TBI and controls

Network deviation index was derived from MRI data acquired after a median 16 days (IQR 13–25 days) and 194 days (IQR 188–210) post-TBI. Fig. 3 shows the total number of deviant white matter connections across the whole brain using a range of z-score thresholds. Participants with TBI have more deviant white matter connections than controls and this effect is consistent across a range of z-score thresholds. Fig. 4a) shows participant level network deviation indices calculated using a z-score threshold of 2.5. Individuals with TBI had significantly higher network deviation index compared to controls in both the subacute ($p = 0.03$, $d = 0.51$) and chronic period ($p = 0.04$, $d = 0.48$) (Fig. 4a) indicating that participants with TBI had a higher number of deviant nodes than controls. There was no difference between network deviation index in the subacute and chronic post-injury period. The proportional change in network deviation index from the subacute to chronic post-injury period was 0.002 (95 % CI -0.1 , 0.01), indicating that at the group level, network deviation index did not significantly change over time; however, network deviation index increased over time for 19 patients while it decreased for 15 patients and remain unchanged in one patient (Fig. 4a). Fig. 4b shows that this relationship is consistent regardless of z-score threshold. Supplementary Fig. 1 shows consistent findings using a higher resolution brain atlas.

3.3. Relationship between network deviation index and acute CT findings and blood biomarkers

Network deviation index in the subacute period was significantly higher among TBI participants with evidence of an acute intracranial lesion on initial head CT (Fig. 5a). Subacute network deviation index correlated with serum GFAP ($r = 0.5$, $p = 0.008$) and NFL ($r = 0.41$, $p = 0.02$) measured in the subacute period. As patients with acute intracranial lesions on initial head CT and blood biomarker elevations tend to be more severely injured, these findings indicate higher levels of network deviation among those with higher injury severity. Chronic network deviation index did not correlate with serum GFAP or NFL measured in the chronic period. (Fig. 5b-c).

3.4. Network deviation index correlations with measures of recovery

Fig. 6 shows the correlation between the proportional change in network deviation index and proportional change in measures of recovery. The proportional change in network deviation index correlated with measures of subjective reporting and functional outcome. There was a positive correlation between proportional change in network deviation index and the BSI and RPQ ($r = 0.46$, $p = 0.03$ for BSI; $r = 0.46$, $p = 0.02$ for RPQ). This indicates that increases in network deviation index from two weeks to six months post-injury correlate with worsening post-concussive symptoms over time. There was also a negative correlation between proportional change in network deviation index and functional outcome as assessed by the Glasgow Outcome Scale-Extended (GOSE) ($r = -0.51$, $p = 0.003$). This indicates that increases in network deviation index over time correlates with less favorable functional outcome over time. We detected no significant association between proportional change in blood plasma concentration of GFAP and NFL with change in outcome scores (BSI, RPQ, and GOSE) from

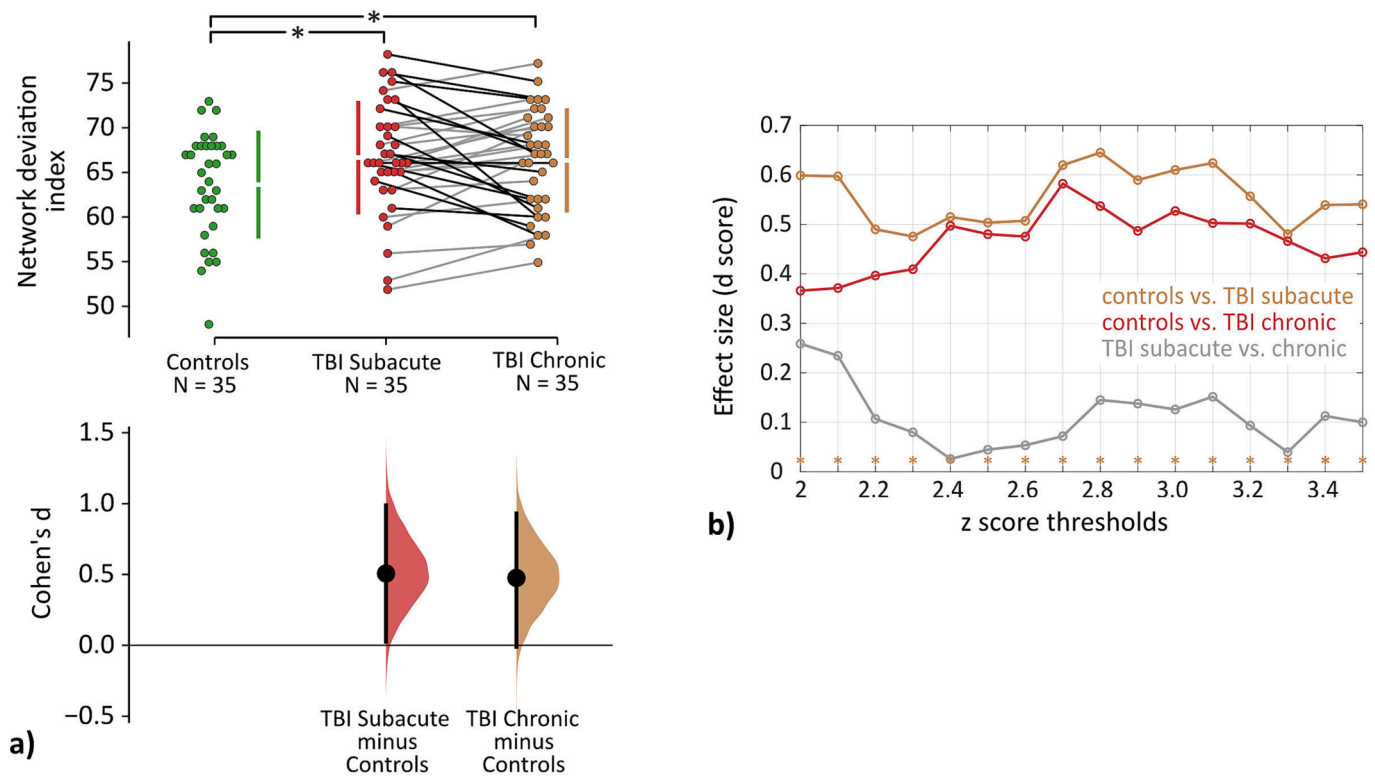


Fig. 4. Participant-specific deviations in whole brain structural connectivity. a) Individuals with TBI have significantly higher structural network deviation index compared to controls both in the subacute (2 weeks: $p = 0.03$, $d = 0.51$) and chronic (6 months: $p = 0.04$, $d = 0.48$) periods after the injury. Results are shown for a z-score threshold > 2.5 . In individual cases, the abnormality load can either increase (gray line), decrease (black line), or remain the same between the subacute and chronic periods. b) These effects are consistent across a range of z-score thresholds. * indicates $p < 0.05$ after FDR correction.

subacute to chronic periods (Supplementary Fig. 2).

3.5. Node deviation index and measures of recovery

Whereas network deviation index estimates the aggregate burden of structural network deviation, node deviation index maps deviation in individual network nodes. To delineate the relationship between node deviation index in the subacute period with recovery status in the chronic period, we utilized elastic net regression. First, we performed a principal component analysis on the total scores for BSI-18, RPQ, and GOSE in the chronic post-injury and extracted the first principal component score, which explained 86 % of the variance in the three measures (Fig. 7a). The model included the first principal component score as the response variable (a measure of global recovery) and 90 features corresponding to the node deviation index of each node in the structural brain network. The correlation between the predicted and actual recovery score was -0.59 (95 % CI -0.77 , -0.31 , $p = 0.001$, Root Mean Squared Error: 0.27, R-squared: 0.23, Fig. 7b). Fig. 7c depicts the feature importance of the subacute structural network nodes for predicting chronic outcome that were selected from 10,000 runs of Monte-Carlo simulation; they corresponded to regions susceptible to neurotrauma (i.e., inferior frontal cortex, temporal pole, occipital lobe) (Oldham and Fornito, 2018; Crossley et al., 2014).

4. Discussion

The main findings in this study are: (1) whole brain structural network deviation index is greater in individuals with TBI compared to healthy controls, (2) in individual TBI cases, the burden of network deviation can increase, decrease, or remain stable from the subacute to chronic post-injury period, (3) network deviation index measured in the subacute post-injury period is significantly higher in those with acute

intracranial lesions visible on CT, (4) network deviation index correlates with serum concentrations of GFAP and NfL in the subacute period, (5) the proportional change in network deviation index correlates with the change in the severity of self-reported symptoms and functional outcome, and (6) subacute network deviation index predicts chronic recovery status after TBI. Collectively, these findings support further investigation of structural network deviations as a patient-specific measure of traumatic axonal injury (TAI).

Network deviation index is a summary score describing deviations in whole brain structural connectivity using a normative modeling framework. Understanding the level of deviation from normal in whole brain structural connectivity may provide a patient-specific estimation of the aggregate burden of TAI. This has important implications for long-term neurocognitive outcomes, given the association between the extent of TAI and the progression of neurodegeneration after moderate-to-severe TBI (Graham et al., 2020). Numerous studies have examined longitudinal changes in white matter microstructural integrity after TBI using diffusion MRI (Wang et al., 2011; Wallace et al., 2018; Palacios et al., 2020; Johnson et al., 2019; Moreira da Silva et al., 2020; Warner et al., 2010; Marquez de la Plata et al., 2011; Dall'Acqua et al., 2017); however, this represents the first study to our knowledge utilizing a normative modeling framework to infer individual level deviations in the structural connectome in a deeply phenotyped cohort of participants with TBI with clinical, blood biomarker, and brain imaging data acquired longitudinally. Moreira da Silva et al. (2020) and Taylor et al. (2020) analyzed group level longitudinal changes in graph theory measures of structural connectivity in patients with primarily mild TBI and found that dynamic changes in connectivity over time are associated with patterns of cognitive recovery. In a cross-sectional analysis, Taylor Neurology (2020) reported a novel, patient-specific measure of injury severity in a cohort of patients with mild TBI by selecting 22 white matter tracts inferred from diffusion tractography and calculating the

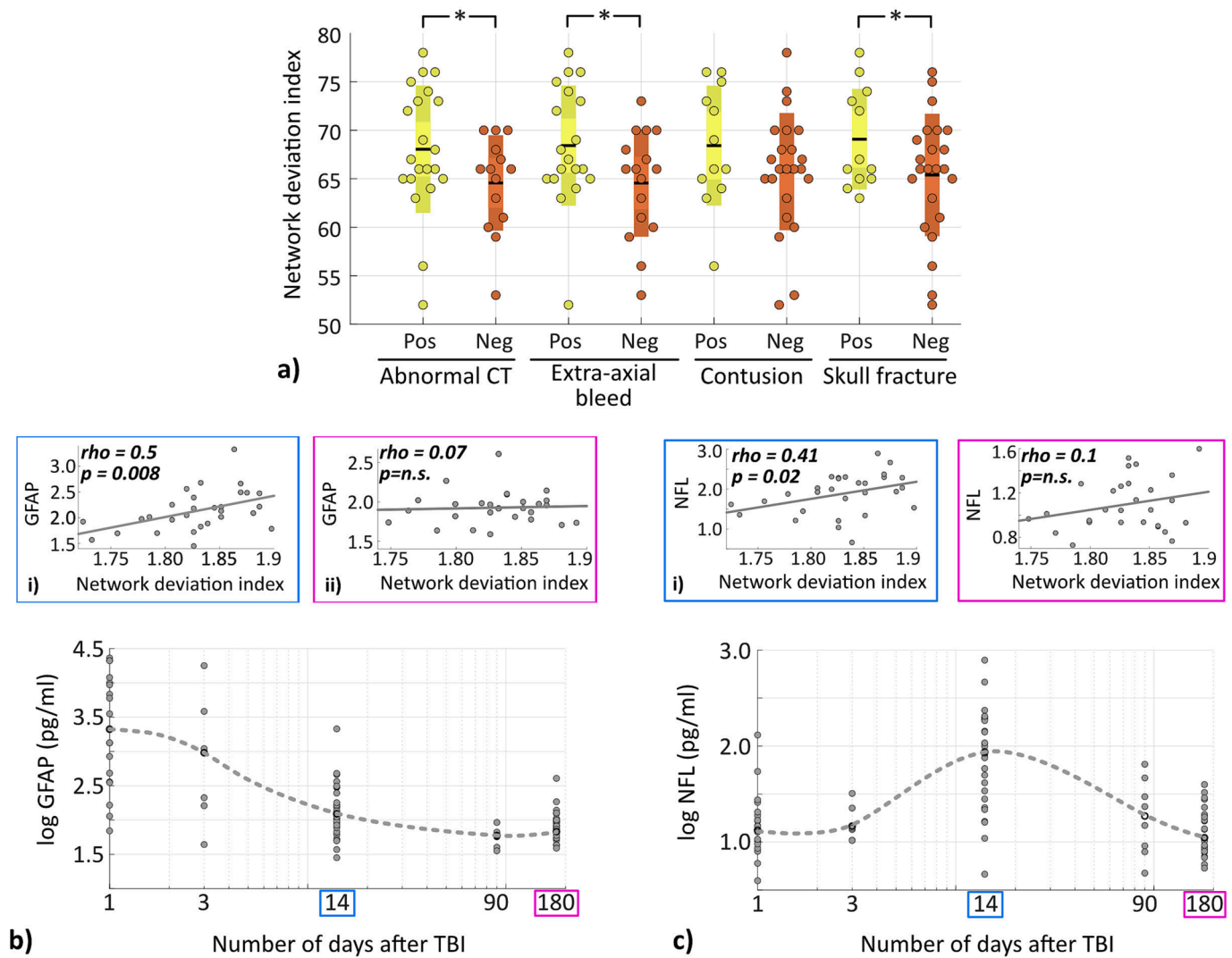


Fig. 5. Network deviation index correlates with independent measures of injury severity. a) Presence of acute intracranial lesions on CT was associated with significantly higher network deviation index than individuals who were CT negative after brain injury (acute post-traumatic lesion on CT: $p = 0.05$, $d = 0.55$; extra-axial bleed: $p = 0.03$, $d = 0.63$; contusion: $p = 0.11$, $d = 0.43$; skull fracture: $p = 0.03$, $d = 0.62$). Plot of trajectory of GFAP (b) and NFL (c) at different time points. These blood biomarkers of injury are expressed highly at different time points after the injury and gradually subside. The inset plots (i) in the blue box shows the correlation between network deviation index with GFAP and NFL in the subacute period, and inset (ii) in the pink box shows the correlation between network deviation index and GFAP and NFL in the chronic period. Increase in network deviation index during the subacute period is correlated with the increase in expression of both blood protein biomarkers in the subacute period (GFAP vs network deviation index: $r = 0.50$, $p = 0.008$; NFL vs network deviation index: $r = 0.41$, $p = 0.02$).

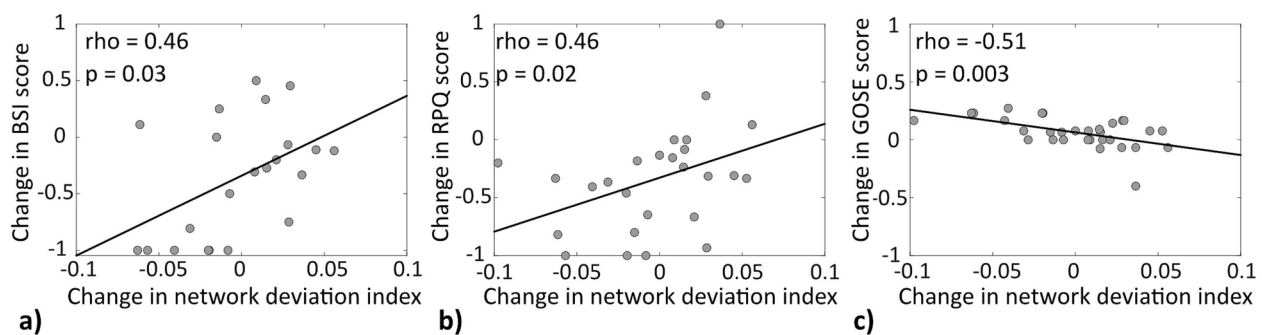


Fig. 6. Network deviation index correlations with outcomes. The proportional change in network deviation index from the subacute to chronic post-injury period correlated with the proportional change in measures of a) subjective psychological symptoms (BSI: $r = 0.46$, $p = 0.03$), b) post-concussive symptoms (RPQ: $r = 0.46$, $p = 0.02$), and c) functional outcome (GOSE: $r = -0.51$, $p = 0.003$). Recovery is indicated by a decrease in BSI and RPQ scores and increase in GOSE score. On proportional change scale, negative values indicate decreases while positive values indicate increase. The correlations indicate that increases in structural network deviation index over time correlates with worsening psychological and post-concussive symptoms as well as functional outcome over time. BSI, Brief Symptom Inventory; RPQ, Rivermead Post-Concussive Questionnaire; GOSE, Glasgow Outcome Scale Extended.

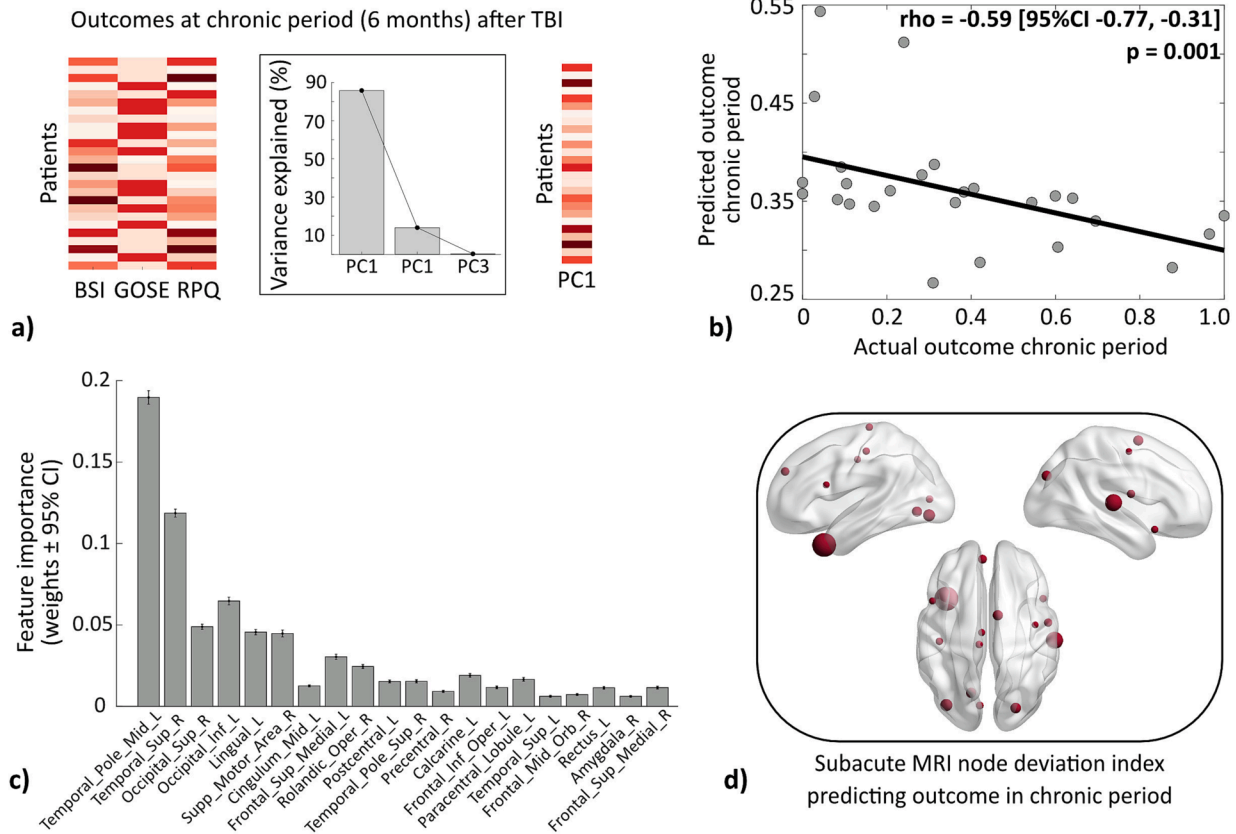


Fig. 7. Subacute node deviation index prediction of chronic outcome. Panel a) shows the results of the principal component analysis performed on BSI-18, GOSE, and RPQ scores in the chronic post-injury period. The score for the first principal component was entered as the response variable in an elastic net regression model with nested cross. Panel b) shows the prediction performance of the optimal model in predicting the chronic post-injury outcome. The predicted performance from the regression model is plotted against the actual change in recovery. The high correlation coefficient between predicted and actual recovery changes $r = -0.59$ [95 % CI -0.77 – -0.31], $p = 0.001$ suggests that subacute node deviation index predicts patient recovery after TBI. Panel c) shows the subacute brain nodes along with their regression coefficients sorted in the order of how frequently they were selected in 10,000 runs of Monte Carlo cross-validation indicating the feature importance of those regions. Panel d) plots the spatial location of these salient brain regions measured in the subacute period predicted chronic outcome. Sphere sizes are drawn to scale with the relative feature importance.

Mahalanobis distance, which represents an aggregate score describing abnormal connectivity. They found that the Mahalanobis distance was able to differentiate patients with mild TBI from healthy controls and that it is correlated with overall cognitive function. This study builds upon these prior studies by reporting on individual level deviations in whole brain structural connectivity in participants with primarily complicated mild TBI from the subacute to chronic post-injury period.

We found that whole brain network deviation index in the subacute post-injury period after mild TBI correlates with concentrations of NfL and GFAP. NfL and GFAP, measured in the acute and subacute period after TBI, predict neurologic outcomes at six months and one year (Lei et al., 2015; Vos et al., 2010). NfL is a marker of axonal injury and is elevated across the spectrum of neurodegenerative diseases (Bacioglu et al., 2016). GFAP is a measure of glial injury; however, one study showed that GFAP was able to distinguish patients with MRI signs of diffuse axonal injury compared with patients with other lesion types (Gugger et al., 2022). Our results are in agreement with prior work, as prior studies demonstrated more robust correlations between NfL and GFAP concentrations with microstructural integrity of the corpus callosum than tau and UCHL1 (Shahim et al., 2020b). For all blood biomarkers examined, we observed peak concentrations during the acute-to-subacute period after TBI followed by declining concentrations in the chronic period. Conversely, network deviation index did not appreciably change between the subacute and chronic periods. This could reflect acute traumatic axonal injury with the acute release of proteins into the bloodstream followed by declining concentrations in

the blood over time, while injured white matter tracts remain disrupted over time as reflected by stability in imaging. How the process of axonal injury and repair is reflected in blood and imaging biomarker profiles during this dynamic process remains to be determined.

We derived a node deviation index for each region defined by the AAL atlas. Node deviation index provides an individual-level measure of regional deviations in structural connectivity from the reference control cohort. To validate this measure, we modeled chronic post-TBI symptoms and functional status using each participant's node deviation index (90 features) from the subacute post-injury period. Using a robust feature selection algorithm, we found that the subacute nodes most informative in predicting chronic outcome map onto areas susceptible to neurotrauma (Sharp et al., 2014). This is an expected finding, as axonal injury severity is likely to be greatest in white matter tracts connected to nodes where visible post-traumatic lesions are commonly seen (e.g., anterior inferior frontal lobe, temporal pole, occipital lobe).

The findings of this work should be interpreted in the context of several limitations. First, patients were scanned at two different, non-uniform time points from the subacute and chronic post-injury period. Structural connectivity changes after TBI likely reflect a dynamic and interactive process of injury, and repair and measurement at two varying time points are unlikely to encapsulate these changes. Second, although at a group level, network deviations did not change over time, it is clear that there is considerable variability where some patients structural network deviations progressed, whereas others became more similar to healthy controls. We did not have the statistical power to

analyze the factors underlying the progressive improvement or deterioration in structural networks. Controls were scanned at a single time point, which also precluded us from establishing a baseline distribution for the trajectory of network deviations over time in uninjured participants. Future studies using longitudinal neuroimaging data from large multicenter studies are necessary to delineate the factors underlying the diverse trajectory of structural network abnormalities. Finally, structural network deviations are inferred from diffusion MRI, which measures water diffusivity. Although primarily thought to reflect axonal injury after TBI, gliosis, atrophy, edema/inflammation, and partial volume effects result in similar effects (Gugger et al., 2022). This concern is partially mitigated by the correlation of network deviation index with canonical blood biomarkers of axonal injury.

We have shown significant multifocal brain structural network deviations in patients with TBI from the subacute to chronic post-injury period. These abnormalities correlate with blood biomarkers of axonal injury as well as recovery after TBI. These findings indicate that structural network deviation measurements warrant further investigations as a biomarker of TAI both at the resolution of the whole brain as well as in individual brain regions. These findings require confirmation in larger longitudinal studies. Nonetheless, this study represents an important step toward developing a non-invasive biomarker that could be used in clinical trials to identify the aggregate and region-specific burden of TAI, monitor the process of post-traumatic epileptogenesis and neurodegeneration, and better patient stratification to track recovery after neurotrauma.

CRedit authorship contribution statement

James J. Gugger: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing. **Nishant Sinha:** Software, Methodology, Formal analysis, Writing – review & editing. **Yiming Huang:** Software, Methodology, Formal analysis, Writing – review & editing. **Alexa E. Walter:** Data curation, Project administration, Writing – review & editing. **Cillian Lynch:** Data curation, Writing – review & editing. **Priyanka Kalyani:** Data curation, Project administration. **Nathan Smyk:** Data curation, Project administration. **Danielle Sandsmark:** Writing – review & editing, Project administration, Supervision. **Ramon Diaz-Arrastia:** Funding acquisition, Project administration, Supervision, Writing – review & editing. **Kathryn A. Davis:** Funding acquisition, Project administration, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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Appendix A. Supplementary data

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Further reading

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